CS241 - Questions for Lab Assignment 5

1 INSTRUCTIONS

(1) This lab **is graded**. Maximum marks: 20
(2) Use the student database that was created in Lab2.
(3) Each question carries 2 marks (1 mark for execution and 1 mark for viva). 10 questions will be chosen arbitrarily from the list below and evaluated.

2 QUESTIONS

Consider the following relations:

Student(snum: integer, sname: string, major: string, level: string, age: integer)
Class(name: string, meets_at: time, room: string, fid: integer)
Enrolled(snum: integer, cname: string)
Faculty (fid: integer, fname: string, deptid: integer)

The meaning of these relations is straightforward; for example, Enrolled has one record per student-class pair such that the student is enrolled in the class.

Write the following queries in SQL. No duplicates should be printed in any of the answers.

You may find it easier to write queries if you use the with clause (or if you create views).

(1) Find the names of all classes that either meet in room 20 AVW or have five or more students enrolled.
(2) Find the number of courses conducted per room.
(3) List all faculty members, showing their id, name and the number of classes they teach. The number of classes of those who teach no classes must be shown as 0.
(4) List all the courses with their names, where they are taught and the number of students enrolled for each. If no students are enrolled, show the number of students as 0.
(5) Find all faculty members who belong to department 20 and whose courses are conducted in room R128.
(6) Find the maximum age of all students of each major
(7) Find the names of students and faculty members whose names contain the string "son".
(8) Find all faculty members who belong to department 20 and whose courses are conducted in room R128. Use a nested query. Do not create views or use a"with" statement.
(9) Find all faculty members who belong to department 20 and whose courses are conducted in room R128. Answer the question by creating a view and using it, without using a nested query.

For the following set of questions, create a new database. The database is shown in Table 1 on the next page:

(10) Find the supervisor of Ravi.
(11) Find the supervisor of the supervisor of Ravi.
(12) Find all the supervisors (direct and indirect) of Ravi.

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Table 1. Employees and Supervisors

person	supervisor
Ravi	Aman
Mary	Sujata
Aman	Devi
Devi	Mary